

Making Sense of Resistance to Agile Adoption in Waterfall Organizations: Social Intelligence and Leadership

Full paper

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Abstract

Organizations wanting to adopt agile software development methodologies often face resistance from development team members with structured development backgrounds. While project management best practices would advocate the need to stifle opposition, recent research suggests resistance may have value in enabling project leaders to respond effectively. Using a qualitative multi-site case study approach, this study elaborates on how scrum masters perceive, interpret, and act upon resistance to enable agile adoption in a waterfall organization. We find that scrum masters need to be more socially intelligent in order to sense the resistance and develop appropriate actions and interventions to enable agile adoption.

Keywords

Iterative Development, Resistance, Social Intelligence, Qualitative Study.

Introduction

After the early excitement of prototyping and wide adoption of waterfall software development methodologies in the 1970's, agile software development is re-emerging as the preferred approach in large organizations (Bose, 2008). Agile represents radical shifts in the software development process away from a structured waterfall approach (Dyba and Dingsoyr, 2008). This shift disrupts team cadence as some resist the new ways of working (Piderit, 2000).

Research on agile adoption primarily focuses on which practices are adopted, how to appropriate practices based on contextual differences, the risks involved, how to empower the agile team, and what is the role of the user (Baliyepally et al., 2009; Cao et al., 2009; Mangalaraj et al., 2009; McAvoy and Butler, 2009; Ramesh et al., 2010). Although these studies provide a basis for examining theoretical frameworks related to how agile projects are adopted in various organizational settings, little guidance exists about the effective mechanisms through which a project leader responds to user resistance towards agile adoption. In this study, we utilize a multi-site case study methodology to examine the following research question:

RQ: How do Scrum Masters react to challenges when faced with resistance to agile adoption in a waterfall organization?

Given the highly social context of agile methodology (Mahadevan and Kettinger, 2015), we find that scrum masters who were successfully able to enable agile adoption, exhibited social intelligence (Lee et al., 2013). Therefore, we believe scrum masters' social intelligence to be a significant factor for future research investigating agile adoption in waterfall organizations. We further build on these results by theoretically explaining how socially intelligent scrum masters were able to perceive and interpret relevant social situational contexts and then devise an appropriate response to developers' resistance based on the information gathered from the interpretation of a social context.

Agile Development, Resistance, and Social Intelligence

Agile development emphasizes intense collaboration among a cross-functional team of developers and customers, to iteratively produce a working software. These iterations, usually 2-4 weeks in length, involves re-prioritizing information requirements so that if changes to the requirements are recognized as the software is being incrementally built, those changes can still be incorporated (Agile Manifesto, 20001). As opposed to agile, waterfall development involves contracting with the customer initially to gather detailed information requirements, which once finalized, are then converted into a final software product by large functional development teams following a sequential plan.

Due to the different underlying philosophies of the two development approaches, agile development adoption can be challenging in traditional waterfall organizations and adjustments are needed depending on the social context (Cao et al., 2009; Fruhling and Vreede, 2006). These challenges include: development team's understanding of the agile methodology, lack of top management and technical support, incompatibility with existing infrastructure and processes, and budget and time constraints (Magalaraj et al., 2009). Other challenges involve collective ownership and adherence to coding standards which requires substantial team member expertise and confidence (Balijepally et al. 2009; Maruping et al., 2009), and scrum masters who need to facilitate and serve the team's needs (McAvoy and Butler, 2009). Notwithstanding the challenges, successful teams that are able to embrace agile approaches have been shown to enhance business value, create a sustainable working rhythm, improve sharing and team learning, and reach higher levels of collective mindfulness, as well as higher software quality and responsiveness (Maruping et al., 2009).

Given the challenges of agile adoption, waterfall organizations often experience resistance from development team members due to the high switching cost incurred when learning radically different agile methods versus following the status quo of waterfall structures (Kim and Kankanhalli, 2009; Polites and Karahanna, 2012; Van Offenbeek and Boonstra, 2013). For example, in traditional development approaches, software customers are less involved and upper management has more control over team processes, while agile methods demand greater customer involvement in project task prioritization and less management control in order to empower the team to make decisions without hierarchal approvals. The customer and the development team may not be comfortable and resist making the decisions. Agile methods demand open communication while traditional methods demand less (Berger and Davies, 2009) leading to resistance from those unused to sharing.

Espousing the need to shift from considering resistance as irrational and dysfunctional, Ford et al. (2008) explore how managers can use the value of resistance in beneficial and meaningful ways. Resistance can offer valuable insights by keeping the conversation going, supporting participation, allowing for legitimizing changes, creating translations and understanding generating scrutiny and counterarguments, offering feedback needed to adjust pace, scope, and the steps of change; improving quality and commitment to change, inviting participation, and providing resources. In the agile context, resistance may best be considered by scrum masters as an informational feedback loop informing them how to take the appropriate actions to ensure agile adoption success. This study examines how managers position themselves to gather insights from resistance.

Ways in which resistance can be responded to include: inaction (i.e., by doing nothing), acknowledgement (i.e., by discussing the issues with resistor), rectification (i.e., by redesigning the old system to fit with resisters' needs), and dissuasion (i.e., by talking the resistor out of their stance) (Rivard and Lapointe, 2012). Although these ways were found to be effective in the context of new technology implementation,

successful scrum masters will also need to be aware of the responses to resistance in order to react appropriately. We build on research about user resistance with accepting new technology implementations to examine how scrum masters respond to users resistance to accept agile method (e.g., Ferneley and Sobreperez, 2006; Klaus and Blanton, 2010; Meissonier and Houze, 2010; Rivard and Lapointe, 2012).

Facilitating the ability to recognize resistance and act upon it appropriately, social intelligence may be a requisite skill. Social intelligence is defined as the ability to recognize, understand, and use emotional information about others that leads to or causes effective or superior performance (Thorndike and Stein, 1937). People with high social intelligence attract others; are friendly, supportive and caring; and generally successful in society, while people with low social intelligence have problems in interpersonal relationships and tend to be misfits in society (Joseph and Laxmi, 2010). Socially intelligent people have the ability to perceive internal states, motives, and behaviors then act optimally on that information (Mayer and Salovey, 1993). It can be developed at the workplace through ‘extroverting’ an individual (Landy, 2005), teaching active listening, and reading body language skills (Groves et al., 2006). The major competencies needed include empathy, persuasiveness, developing others, group management, networking, negotiating, and oral communications (Boyatzis and Ratti, 2009).

Research Methodology

Due to the limited understanding of the process of responding to resistance in software development methodology adoption, we take an exploratory approach using a case study methodology (Eisenhardt, 1989; Yin, 1994). A case study methodology is appropriate for the context of our paper because it allows us to integrate interviewee’s views thereby providing us with insights into behaviors that might otherwise be difficult to find (Whitley, 1995). We selected a Fortune 500 multinational service organization, ParentCo (a pseudonym), headquartered in the United States, because the entire company has been following a traditional waterfall development process for years and has been attempting to move toward greater agile method adoption. ParentCo is a global organization operating for over 40 years, in more than 200 countries, with over 4,000 technology employees and over \$10 billion in revenues. ParentCo has three major releases of hundreds of new and modified software products per year. ParentCo recently announced the need to improve its ‘speed-to-market’ and systems development in two of its six divisions: the Retail division and the Logistics division. Our data collection began in these two divisions approximately one year after initial adoption with about 8% of the releases developed following agile approaches. Both divisions hired the same national training and consulting firm to mentor and coach employees on how to implement these practices.

Data Collection

We collected qualitative data over a three month period from January 2013 to March 2013. First, four background interviews with scrum masters at companies expert in practicing agile methodologies were conducted to help us develop an interview protocol. At the Retail and Logistics divisions of ParentCo, we gathered data using semi-structured and structured face-to-face interviews with scrum masters to better understand the issues related to adopting agile practices. In total we conducted interviews with 12 project managers, 3 developers, 3 testers, and 1 analyst. Each interview lasted between 45-70 minutes. Hand written notes were also taken during the interviews which were then added to the transcribed tapes used during interviews to generate a total of 278 pages of notes. The interviews focused on several dimensions of agile development including the characteristics of activities, factors that influence the decision to adopt a new approach, and the project management issues. We also examined software development process documentation to gain an in-depth understanding of ParentCo.’s existing waterfall development process.

Data Analysis

As data was being collected, we started to analyze the major themes that were then used to guide future interviews. This allowed us to refine our protocol as interviews progressed and probe relevant questions that could provide deeper insights for our research. Data collection resulted in a presentation of various facets of agile adoption, such as the scrum masters’ leadership strategy, governance process, budgeting issues, team management issues, and personality fit with new practices. At this time, the theme of resistance, social intelligence, and scrum master responses was beginning to emerge as crucial factors in

both divisions. The second phase of our data analysis was more elaborate and focused on the topics of resistance and social intelligence. We used open coding techniques (Strauss and Corbin, 1990) to code qualitative data by categorizing it into separate units of meaning and assigning each category a conceptual code. At the end of the process, we generated 142 interview text segments that produced 104 concepts reflecting concerns of agile adoption, project management, team member resistance, governance processes, and social intelligence, among others. We next sorted and clustered the concepts by comparing and contrasting them with each other to determine commonalities and differences between them and to generate broader categories (Miles and Huberman, 1994). We ended data collection when theoretical saturation was reached (Orlikowski, 1993; Strauss and Corbin, 1990) and when no new category or linkages among the categories emerged during the coding. Following axial coding procedures (Strauss and Corbin, 1990), we determined the essential paradigmatic properties of each category in terms of whether it embodied context, condition, interactional strategy, or the consequences of our central phenomenon of resistance and reactions. The first category reflected the organizational, project, and individual environmental contexts from which scrum masters realize resistance from team members. The second category reflected how scrum masters sensed resistance using social intelligence. The third category reflected the consequences of using these mechanisms on the extent of agile adoption.

Results

We summarize in Figure 1 the contexts where stakeholders realize that influence how scrum masters recognize resistance using socially intelligent means. Finally we illustrate how resistance recognition leads to specific responses. The categories and concepts identified here are not exhaustive, but are those that emerged as salient at ParentCo. Future research should develop, refine, and validate this model.

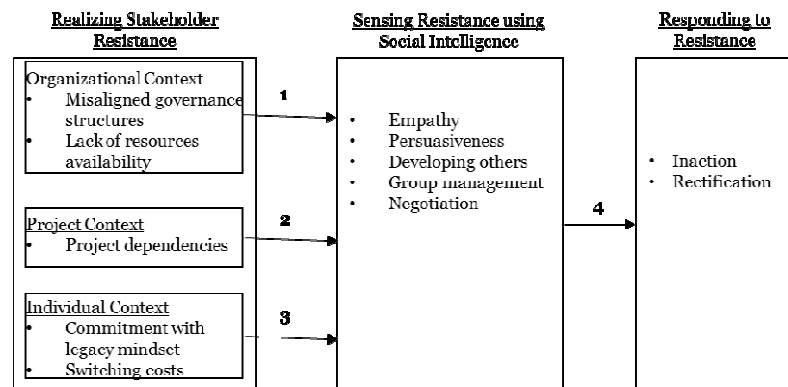


Figure 1. The Process of Responding to Resistance in Agile Development Adoption.

Process of responding to resistance

Here, we describe the process of how scrum masters responded to resistance. The realization of resistance from stakeholders started the process of identifying resistance behaviors to induce either a lack of action or rectification of stakeholder resistance when pursuing agile adoption.

Realizing stakeholder resistance

To respond to development stakeholder resistance, it was crucial for the scrum master to first be aware of resistance and the contextual reasons stakeholders resist agile development practices. We categorized stakeholders' resistance behaviors in three sub-categories namely resistance due to the organizational context, project context, and individual context.

Organizational Context

Misaligned governance structure. As a global, multi-division, company with regulatory compliance requirements, ParentCo established, enforced, and dogmatically-applied formal governance processes and procedures on all facets of the business including software development. The governance processes

spelled out detailed activities mandated to ensure all employees consistently met oversight rules. However, agile development practices encourage self-driven, creative, collocated work environments with short-term planning in budgets, resources, and activities. At ParentCo scrum masters explained how complying with misaligned governance processes (e.g., budgeting and scope planning, accounting documentation and audit trails, time tracking systems) made agile development challenging to achieve:

“The biggest risk we have with that is again, dealing with linear interface of the governance people and you know the folks are doing agile and are saying governance is getting our way, the governance people are saying, you guys are bunch of range cowboys.” Scrum master 7

Lack of resources availability. ParentCo is a hierarchal structure where employee reporting is based on relatively siloed functional units. With recent downsizing, most employees have a full workload including both project and operational tasks to accomplish. People on development teams are assigned to multiple projects under waterfall-structured work plans with buffer time built in. In contrast, agile development approaches need fully dedicated team members representing all development roles (i.e., customer, analyst, designer, developer, tester, etc.) working together closely to iterate through development backlogs. The cross-functional team means resources across functional units are needed on a full-time basis. Team members rotate roles and pick up the slack for other members of their team. At ParentCo the lack of dedicated cross-functional resources was cited as one of the largest roadblocks to agile team work:

“The other challenge we have that keeps us from being purely agile is the resources on the project are not 100% dedicated to this project. They have to divide their time, and oddly enough they divide their time with another project, which is waterfall.” Scrum master 11

Project Context

Project dependencies. Being a large multinational company, ParentCo’s information systems comprise several monolithic comprehensive computer programs decades old. Furthermore, having grown through acquisitions over the last fifty years, various inherited systems were integrated into existing ones. These complexities impede the ability to modularize and isolate projects so agile teams can quickly develop architecturally-sound new software products or enhancements. At ParentCo team members complained about the overwhelming challenges the legacy systems environment caused in using agile development practices. As one scrum master put it:

“The problem is dedicated resources and interconnected environments, i.e., products that impact several other products.” Scrum master 14

Individual Context

Commitment to legacy mind set. At ParentCo, all system development stakeholders (i.e., developers, designers, customers, etc.) are well trained and indoctrinated in organizational structured processes including the waterfall development process. However, failing to let go and remain committed to non-agile processes threatened to block the transition to following agile practices. Stakeholders were sceptical, stubborn old technologists, who needed to step out of their comfort zone and spend time getting on board with the new process. At ParentCo, risk-averse employees got set in their ways and were comfortable with structured waterfall development processes that had worked in the past. Many newcomers to agile projects had no desire to move away from the status quo, they were inflexible and keep thinking with a mentality of waterfall procedures. This resistance also include the customer who would not prioritize task backlogs, because they were used to the waterfall process of tossing requirements over the wall for development to take over:

“They’ve [team members] demonstrated success using that methodology in the past and they’re comfortable with what they know and they’re somewhat risk-averse, so there’s no desire to change the status quo.” Scrum master 3

Switching Cost

a. Lack of trust and threat to status or power. Agile teams are self-directing and team-focused which necessitates equality and contribution by everyone. In traditional organizations, most employees, especially in the development ranks, prefer some level of recognition as the go-to person or expert and star of the group (Moe et al, 2012). Agile teams require team members and stakeholders to set aside

personal agendas and egos to focus on the success of the collective. For agile methods to succeed, stakeholders must trust others in making decisions without controlling but enabling their actions. At ParentCo, technology management often wants to lock the customer down with waterfall-type sign-offs because the customer has accused the technology side of being difficult to work with in the past. Top executives from both sides want everything decided at the start by establishing a hard set of system requirements and then treating it like a contract. However, with agile methods, requirements are not known up front and change continuously through reprioritization. Agile environments require flexibility and trust. Thus, top management resists allowing teams to work agilely complaining that it is too open for changes, they want a contract not collaboration:

“It’s just so unfortunate we can’t have that trust...And you feel there’s no reason for them [top management] to not trust us because we aren’t stupid.” Scrum master 13

Agile development requires focusing on team needs and activities over the individual. Each person is one among the group striving to help everyone move toward a successful project outcomes. With team ownership and self-direction, scrum masters, project managers, and top executives may find letting go of their hierarchical control difficult (Boehm and Turner, 2005; Conboy et al., 2011). Agile practices mean letting go of control of your domain and playing on a team, which threatens executive control:

“It would be hard for a scrum master that’s been a command and control kind of person, used to that, to all of the sudden relinquish that control, give that control to the team and sit back and be more of a mentor.” Scrum master 2

b. Discomfort with agile interactions. Because ParentCo. has been developing software using the structured waterfall methodology for a long time, many of the developers develop working habits and behaviors in synch with structured methodologies. At ParentCo., we often heard how software developers prefer isolated and non-social working environments. Yet agile methodologies require team members to actively collaborate with other team members on a day to day basis. Such an environment may become a cause of discomfort for introverted developers (Boehm and Turner, 2005; Conboy et al., 2011; Mahadevan and Kettinger, 2015). As one scrum master put it:

“You need normally the more mature people to deal in because you are in close proximity you need people who communicate well with others. I mean there are those brilliant development types who have to be given exactly what to do then they do it and they come back ask for a candy, that won’t work in agile” Scrum master 5

In summary, while corporate governance and compliance process may be a burden, resource issues were inconvenient, and modularization and isolation may be not be possible, some scrum masters used these resistance factors as informational input for figuring out how to rectify the situation with procedure workarounds and exemption approvals along with knowledge sharing and expectation management (i.e., keeping customers apprised of work delays due to resources constraints) as well as bringing architectural staff on project teams. Some scrum masters with stronger sensing abilities were able to ferret out which employees were better suited for agile work environments as well as negotiate and communicate with stakeholders to gain buy-in with agile practices and sense when their teams and stakeholders are struggling and use this as an opportunity to develop others. Highly aware scrum masters perceive situations lacking trust and threatening power and of general discomfort in order to intervene and work with stakeholders to move past personal issues to focus on solutions. These highly sensitive scrum masters were able to realize resistance as a source of information and enabled agile practices to flourish.

Sensing resistance with social intelligence

Social intelligence is the ability to identify and use emotional information about others in effective ways leading to superior outcomes (Boyatzis and Ratti, 2009). Our data showed that scrum masters who were in a better position to find informational value in resistance also exhibited significant levels of social intelligence i.e. empathy, persuasiveness, developing others, group management, and negotiation skills.

Empathy. Empathy enables people to understand others’ emotions, perspectives, and take an active interest in what is important to them. For example, it helps scrum masters identify when resisters feel uncomfortable with the close interactions among stakeholders in agile project environments. Empathetic leaders can carefully select team members that will thrive in an individualistic, self-directed, success-

driven, goal-oriented team collaboration situation. Agile development project teams focus on team accomplishment, continuously refactoring code, using design patterns, writing software tests, leveraging existing code, focusing on usability, and coding in any language. Team members need to be highly-confident, socially-oriented, emotionally-capable workers who are proud of their team contributions, and proficient in using technology tools and agile-type processes. At ParentCo we often heard how software developers may not fit with this type of working environment:

“Try to fit that process with the kind of resources who are going to be really good at that kind of work anyway and get the most out of your best people and allow them to work in a way that probably they’re more suited to work....Look at your work force and try to help them understand, like, well you’re here for this purpose, so we’re going to put you over here to suit your goal and your personality better.” Scrum master 4

Persuasiveness. Persuasiveness helps people to exert a powerful influence over others. It provides scrum masters the ability to reason with others, compromise, and discuss legitimate options. Through these interactions resistors develop trust in the process which in turn helps them to be more accepting of agile methodologies. At ParentCo we found scrum masters took persuasive approaches to addressing resistance:

“I guess I pushed it a lot. Management bought into it. And once they bought into it, they were skeptical but they still supported me in it. And then once they saw the amount of work that could be done.” Scrum master 6

Developing others. Leaders engaged in developing others will act as a role model, reinforce the importance of continuous learning, build sustainable process for staff development, reinforce shared values, and leverage problems as opportunities for growth. The need to develop others helps a scrum master know when resistance stems from lack of knowledge and experience with agile development practices.

“In the beginning you may have had to do some handholding and facilitation so that people don’t misunderstand each other. You know, there’s some mediation going on.” Scrum master 2

Group management. Leaders with strong group management skills understand the need to ensure collective objectives are communicated, explain project roles and accountability, and specify clear rules of engagement. These leaders will want to build smaller close knit heterogeneous teams through team building and group processing activities. Scrum masters’ in possession of group management competencies will be in a better position to plan team resources in such a way that team members are not doing monotonous tasks. This keeps the creativity of the team members alive which is an important aspect of software development. A couple of scrum masters put it:

“... you need to make sure that people are given different features and they are not doing the same thing over a period of time...you need to plan in such a way that you are changing resources from time to time and getting fresh air and at the same time keeping the knowledge base within the project team and making sure everyone understands the end result or end goal of this project, as well as the timelines and making sure their activities are aligned to meet the end result is important.” Scrum master 8

Negotiating. Negotiation is a dialogue and potential compromise among parties encouraging a shared understanding or problem resolution by producing an agreed upon course of action or bargaining for certain outcomes. Scrum masters possessing strong negotiating power allows them to take care of the barriers that team members may face when developing software agilely:

“And I did it over and over and over and over again until they listened... then we started doing it. They paid for training, we started doing it, and then it was more of an okay, we’re going to try this because we do see there might be some value in it. So we tried it and then the more we got better at it.” Scrum master 2

Through strong social intelligence skills of empathy, persuasions, developing others, group management, negotiating, and oral communications, those leading agile development projects in environments with resistance are better positioned to sense the value in resistance behaviors and act on them accordingly.

Responding to resistance

Based on our observations at ParentCo and as discussed above, socially intelligent scrum masters reacted to resistance via one of two ways: inaction or rectification. Inaction indicates a lack of action which could occur due to unawareness, deliberate ignorance, or impotence. Rectification defines corrective responses intended to alter the system, the environment, or the stakeholders themselves by congruent or non-congruent means. We did not find evidence of acknowledgement and dissuasion responses to resistance. Therefore we did not include them in our process model.

Relationships between the contexts and sensing resistance value with social intelligence (Arrow 1, 2, and 3: Figure 1)

An examination of the environmental context of ParentCo and its Retail and Logistics divisions illustrated how various contextual differences may play a role in the process of responding to resistance. ParentCo is an organization with a waterfall structured governance oversight process which instills the necessity of inflexible process checklists, hierarchical reporting structures, and layers of approvals into employee mindsets and operating procedures. For example, scrum master 7 noted

“when you say what are the hardest things about making the transition is most companies are, their governance processes are linear and that’s the biggest issue” Scrum Master 7

At ParentCo, individual challenges involved employees with legacy waterfall work habits and capabilities, suggesting that for these individuals, switching costs would be high for reasons including threat to the status of power within the team, wanting to maintain status quo and discomfort with frequent social interactions with other team members (Chan and Thong, 2009; Mahadevan and Kettinger, 2015)

All levels and types of resistance could be found in the interviews at ParentCo, however, how scrum masters went about dealing with these complaints was dramatically different. Scrum masters fell into two divided approaches: able to recognize value in stakeholder comments of resistance and use these comments as input to discerning the actions to take in response versus unable to take responsive actions, expressing the team members were fostering unreasonable obstacle or barriers intent on stopping agile adoption. Agile leaders failed to see the value in complaining but saw complaining as something to stifle. Consistent with recent discussion in the management literature, Ford et al. (2008) suggest a change agent’s job (scrum master leading agile adoption) must include responsibility for gaining buy-in by the team along with the tactics of implementation. This responsibility includes owning the ability to sense and respond to the root reasons of apparently resistive behaviors. The leader must bring conversations to the forefront and engage in responsive actions needed to maintain and improve the team’s relationships. This requires sensemaking and social awareness. The goal is to not suppress but engage contributions to change through understanding the value of resistance.

Relationships between recognizing resistance using social intelligence and responding to resistance (Arrow 4: Figure 1)

Given the apparent successful ability of many scrum masters to sense value in resistance and respond accordingly, we explored how these leaders were able to achieve these results. We found those who exhibited social intelligence i.e. empathy, persuasiveness, group management skills, negotiation skills, and ability to develop others were in the best position to respond to resistance in ways that led to greater agile adoption. For example, at both the Retail and Logistics divisions, scrum masters with the desire to develop the skills and talents of others are able to see through the comments (i.e., surface level resistance) to tease out that the root of the problem was really the person’s lack of understanding of agile methods.

Scrum masters who are empathic and have strong group management skills understand the need to enable team members to work together as a group by identifying the appropriate members to participate on the team. Scrum masters with the ability to reason with others, compromise, and discuss legitimate options help establish trust among team members espousing resistant views. In general, social intelligence provides scrum masters the innate capacity to: perceive when resisters feel threatened because they are losing their status or power, understand when resistance involves an individual putting themselves above the team, identify when resisters feel uncomfortable with the close interactions among

members, understand when conformance though inaction is a necessary response to resistance, adopt essential workarounds to misaligned structures, and know when resistance stems from a lack of resources.

Conclusion

Our research is motivated by the primary research question: “How can Scrum Masters react to challenges when faced with resistance to agile adoption in a waterfall organization?” The challenges faced by scrum masters indicate that agile development adoption entails a significant investment of time and effort on a scrum master’s behalf, and scrum masters should not assume that teams resistance is fostering unreasonable obstacle or barriers intent on stopping agile adoption but recognize value in stakeholder comments of resistance and use these comments as input to discerning the actions to take in response versus unable to take responsive actions. Since social intelligence can be developed, organizations will need to ensure that scrum masters are trained to acquire social intelligence to manage and develop relationships with those resisting agile on their team.

REFERENCES

- Balijepally, V., Nerur, S. P., and Mahapatra, R. K. “IT Value of Software Development: A Multi-theoretic Perspective,” In proceedings of the *American Conference on Information Systems AMCIS*, 2007 p. 294.
- Barlow, J., Giboney, J., Keith, M., Wilson, D., Schuetzler, R., Lowry, P., and Vance, A. 2011. “Overview and guidance on agile development in large organizations,” *Communications of the Association for Information Systems* (29:2), pp. 25-44.
- Berger, H., and Beynon-Davies, P. 2009. “The utility of rapid application development in large-scale, complex projects,” *Information Systems Journal* (19:6), pp. 549-570.
- Boehm, B., and Turner, R. 2005. “Management challenges to implementing agile processes in traditional development organizations”. *Software, IEEE* (22:5), pp.30-39.
- Bose, I. 2008. “Lessons learned from distributed agile software projects: a case-based analysis,” *Communications of the Association for Information Systems* (23:1), pp. 619-632.
- Boyatzis, R. E., and Ratti, F. 2009. “Emotional, social and cognitive intelligence competencies distinguishing effective Italian managers and leaders in a private company and cooperatives,” *Journal of Management Development* (28:9), pp. 821-838.
- Cao, L., Mohan, K., Xu, P., and Ramesh, B. 2009. “A framework for adapting agile development methodologies,” *European Journal of Information Systems* (18:4), pp. 332-343.
- Chan, F. K., and Thong, J. Y. 2009. “Acceptance of agile methodologies: A critical review and conceptual framework”. *Decision Support Systems*, (464), pp. 803-814.
- Conboy, K., Coyle, S., Wang, X., and Pikkariainen, M. 2011. “People over process: Key challenges in agile development”. *IEEE Software* (28:4), pp. 48-57.
- Dybå, T., and Dingsøyr, T. 2008. “Empirical studies of agile software development: A systematic review,” *Information and Software Technology* (50:9), pp. 833-859.
- Eisenhardt, K. M. 1989. “Building theories from case study research,” *Academy of Management Review* (14:4), pp. 532-550.
- Ferneley, E. H., and Sobreperes, P. 2006. “Resist, comply or workaround? An examination of different facets of user engagement with information systems,” *European Journal of Information Systems* (15:4), pp. 345-356.
- Ford, J. D., Ford, L. W., and D'Amelio, A. 2008. “Resistance to change: The rest of the story,” *Academy of Management Review* (33:2), pp. 362-377.
- Fruhling, A., and Vreede, G. J. D. 2006. “Field experiences with eXtreme programming: developing an emergency response system,” *Journal of Management Information Systems*. (22:4), pp. 39-68.
- Groves, K. S., Pat McEnrue, M., & Shen, W. 2008. “Developing and measuring the emotional intelligence of leaders,” *Journal of Management Development*. (27:2), pp. 225-250.
- Joseph, C., and Lakshmi, S. 2010. “Social Intelligence, a Key to Success,” *IUP Journal Of Soft Skills* (4:3), pp. 15-21.
- Kim, H. W., and Kankanhalli, A. 2009. “Investigating user resistance to information systems implementation: A status quo bias perspective”, *MIS Quarterly* (33:3), pp.567-582.

- Klaus, T., and Blanton, J. E. 2010. "User resistance determinants and the psychological contract in enterprise system implementations", *European Journal of Information Systems* (19:6), pp.625-636.
- Kumar, N., Rose, R., and Subramaniam 2008. "The bond between intelligences: Cultural, emotional, and social," *Performance Improvement* (47:10), pp. 42-48.
- Landy, F. J. 2005. "Some historical and scientific issues related to research on emotional intelligence", *Journal of Organizational Behavior* (26:4), pp. 411-424.
- Lee, H., Park, J., and Lee, J. 2013. "Role of leadership competencies and team social capital in IT services," *Journal of Computer Information Systems*, summer, pp. 1-11.
- Mahadevan, L., Kettinger, W. J., & Meservy, T. O. 2015. "Running on Hybrid: Control Changes when Introducing an Agile Methodology in a Traditional "Waterfall" System Development Environment". *Communications of the Association for Information Systems* (36:1), pp. 5.
- Mangalaraj, G., Mahapatra, R., and Nerur, S. 2009. "Acceptance of software process innovations—the case of extreme programming," *European Journal of Information Systems* (18:4), pp. 344-354.
- Maruping, L. M., Venkatesh, V., and Agarwal, R. 2009. "A control theory perspective on agile methodology use and changing user requirements," *Information Systems Research* (20:3), pp.377-399
- Maurer, R. 2006. "Resistance and change in organizations," *The NTL handbook of organization development and change: Principles, practices and perspectives*, pp. 121-138.
- Mayer, J. D., and Salovey, P. 1995. "Emotional intelligence and the construction and regulation of feelings," *Applied and Preventive Psychology* (4:3), 197-208.
- McAvoy, J., and Butler, T. 2009. "The role of project management in ineffective decision making within Agile software development projects," *European Journal of Information Systems* (18:4), pp.372-383.
- Meissonier, R., and Houzé, E. 2010. "Toward an 'IT Conflict-Resistance Theory': action research during IT pre-implementation," *European Journal of Information Systems* (19:5), pp.540-561.
- Miles, M. B., and Huberman, A. M. 1994. *Qualitative data analysis: An expanded sourcebook*. Sage.
- Moe, N. B., Aurum, A., and Dybå, T. 2012. "Challenges of shared decision-making: A multiple case study of agile software development". *Information and Software Technology* (54:8), pp. 853-865.
- Nerur, S., Mahapatra, R., and Mangalaraj, G. 2005. "Challenges of migrating to agile methodologies." *Communications of the ACM* (48:5), pp. 72-78.
- Orlikowski, W. J. 2000. "Using technology and constituting structures: A practice lens for studying technology in organizations," *Organization Science* (11:4), pp.404-428.
- Piderit, S. K. 2000. "Rethinking resistance and recognizing ambivalence: A multidimensional view of attitudes toward an organizational change," *Academy of Management Review* (25:4), pp.783-794.
- Polites, G. L., and Karahanna, E. 2012"Shackled to the Status Quo: The Inhibiting Effects of Incumbent System Habit, Switching Costs, & Inertia on New System Acceptance," *MIS Quarterly*(36:1),pp.21-42.
- Rahim, M. A. 2014. "A structural equations model of leaders' social intelligence and creative performance." *Creativity and Innovation Management* (23:1), pp. 44-56.
- Ramesh, B., Cao, L., and Baskerville, R. 2010. "Agile requirements engineering practices and challenges: an empirical study," *Information Systems Journal* (20:5), pp.449-480.
- Riggio, R. E. 2010. "Before Emotional Intelligence: Research on Nonverbal, Emotional, and Social Competences," *Industrial and Organizational Psychology* (3:2), pp.178-182.
- Rivard, S., and Lapointe, L. 2012. "Information technology implementers' responses to user resistance: nature and effects," *MIS Quarterly* (36:3), pp.897-920.
- Sarker, S., and Sarker, S. 2009. "Exploring agility in distributed information systems development teams: an interpretive study in an offshoring context," *Information Systems Research*, (20:3), pp.440-461.
- Seal, C. R., Boyatzis, R. E., and Bailey, J. R. 2006. "Fostering Emotional and Social Intelligence in Organizations," *Organization Management Journal* (3:3), pp.190-209.
- Strauss, A., and Corbin, J. 1994. "Grounded theory methodology," *Handbook of qualitative research*, pp. 273-285.
- Thorndike, R. L., and Stein, S. 1937. "An evaluation of the attempts to measure social intelligence," *Psychological Bulletin* (34:5), 275.
- Van Offenbeek, M. A., Boonstra, A., and Eseryel, U. Y. 2013. "The Dynamic Interplay among Institutional Logics Influencing Hospital IT Governance," *European Journal of Information Systems* (22:4), pp.434-454.
- Whitley, B. 1995. *Principles of Research in Behavioral Science*. Mountain View, CA, Mayfield.
- Yin, R. K. 1994. *Case Study Research—Design and Methods*. Applied social research method series. Sage, London.